

MELISSA N. HORGER, PHD

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TECHNICAL SKILLS

- R, Python
- Data manipulation and visualization
- Neuroimaging (pediatric EEG & MRI)
- Scientific and grant writing
- Team management
- Research presentations
- Statistical analysis
- Budget planning and management
- Industry and academic collaboration
- Mentorship and training
- Qualitative and quantitative methods
- Evidence synthesis

WORK EXPERIENCE

Postdoctoral Researcher | University of Massachusetts Amherst July 2023 - Present

- Managed a team of 15+ full time staff and undergraduate students to effectively recruit and collect physiological, behavioral, and survey data from pediatric populations and their caregivers
- Organized training opportunities for graduate students and staff in EEG methodology and multilevel modelling
- Launched 2 large federal grants (combined \$1M+) aimed at discovering the impact of sleep in memory consolidation and neural maturation during formative stages of development
- Monitored longitudinal data collection (of over 100 participants thus far) ensuring adherence to quarterly goals
- Spearheaded a validation study with industry partners and disseminated results to in conferences and symposia
- Lead lab-wide communications with suppliers for purchasing, troubleshooting, and updating equipment

Postdoctoral Researcher | Temple University June 2021- July 2023

- Collaborated with international companies to customize physical and digital study materials
- Wrote experimental protocols for the Institutional Review Board in compliance with ethical guidelines
- Oversaw the collection of biweekly data from 110 infants and their families resulting in a repository of over 600 videos. Videos were semi-automatically coded using a computer vision algorithm
- Performed advanced decomposition of EEG data (Principle Component Analysis; ERPs) and ran multilevel models to quantify the impact of individual-level factors on neural plasticity in the sensorimotor cortex in a sample of 60

Adjunct Instructor | Temple University & College of Staten Island Aug 2018 - May 2021

- Taught 5 sections of introductory and upper-level Psychology courses to classes of 22-49 students
- Designed semester-long group research projects with incremental goals and tangible final products
- Balanced a diverse set of student needs while maintaining high standards by writing clear and comprehensive syllabi and grading rubrics

Graduate Research Assistant | Research Foundation, CUNY Aug 2016- June 2021

- Designed a novel method for studying infant sleep in the home with applications for basic and applied research
- Analyzed data for publication via methods such as cluster analysis, time series analysis, and logistic regressions
- Mentored 26 undergraduates on a range of research projects, 5 of whom completed undergraduate theses
- Maintained substantial volumes of video data (>1250 hours) stored in an online repository
- Received training in quantitative and qualitative methods and analyses through formal coursework and assignments
- Conducted literature reviews and synthesized peer reviewed research on interdisciplinary topics

GRANTS, PUBLICATIONS, AND PRESENTATIONS

- Received ~\$200K in grants and awards during my doctoral and post-doctoral studies, including a >\$100K National Research Service Award (NRSA) from the NIH
- Published **12** peer-reviewed articles (including 6 as first author) and served as a reviewer for **7** journals
- Presented at **27** professional conferences (e.g., Annual Meeting of the Associated Professional Sleep Societies), submitted and received acceptances for **31** posters, and organized and/or presented at **7** symposia
- Gave **6** invited talks organized by other departments, universities, or industry collaborators

EDUCATION

PhD, MA | Psychology Graduate Center, CUNY

Aug 2016 - Sept 2021

BA | Psychology Millersville University

Aug 2009 - May 2013